PROJECT REPORT

CLOUD ARCHITECTURE DESIGN – AWS CLOUD PROJECT

PROJECT TITLE

[Enter the approved project title]

Submitted by: [Student name(s) and register number(s)]

Programme / Branch / Year: [Details]

Course Code and Name: BCSE355L – Cloud Architecture Design

Institution: Vellore Institute of Technology, Vellore

Course Instructor / Guide: Dr. Priya V

Team Number: [Team number / Individual project]

GitHub Repository: [Repository URL]

Date of Submission: [DD/MM/YYYY]

Document Preparation Instructions

Use this template for the AWS Cloud Architecture Design project. Replace every bracketed prompt with project-specific content, remove instructional notes before submission, and ensure that all diagrams, tables, screenshots, results, and references are clearly labelled. Do not claim implementation, measurements, or results that have not actually been produced.

- Abstract: 200–300 words.
- Literature survey: 15 research papers, preferably published during 2023–2026, from recognised scholarly publishers/indexed journals.
- Every student must independently analyse their assigned papers and identify methods, advantages, limitations, research gaps, and possible improvements.
- Include both mandatory architecture diagrams: AWS Cloud Architecture and Complete System Architecture.
- Include dataset details where a dataset is used; if no dataset is applicable, explain why.
- Document the AWS services planned/used, GitHub workflow, team contribution, implementation progress, and evidence.
- Formatting: Times New Roman 12 pt; headings 14–16 pt bold; 1.5 line spacing; 1-inch margins; page numbers at bottom centre. Use APA style for research-paper references.

Abstract

[Write 200–300 words covering the problem statement, existing challenges, proposed solution, main AWS services, methodology, current implementation status/results, and conclusion. For Phase I, clearly distinguish planned work from completed work.]

Keywords

[Provide 4–6 keywords, e.g., AWS, cloud architecture, scalability, serverless computing, cloud security, monitoring.]

Table of Contents

In Microsoft Word, use References → Table of Contents to insert an automatic table of contents after applying the heading styles.

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1. Introduction

1.1 Background

[Introduce the application domain and explain why a cloud-based solution is relevant.]

1.2 Problem Statement

[State the specific problem, affected users/stakeholders, and the shortcomings of the current approach.]

1.3 Motivation and Significance

[Explain the practical need and expected value of the project.]

1.4 Scope

[Define what is included and excluded in the project.]

1.5 Report Organisation

[Briefly describe the content of each chapter.]

2. Literature Survey / Related Work

Review 15 relevant research papers. Prefer papers from 2023–2026 and sources such as IEEE, Springer, Elsevier, ACM, Wiley, MDPI, Nature, or Scopus-indexed journals. Explain how each paper relates to the project rather than listing abstracts.

Paper ID	Full citation / DOI / URL	Problem addressed	Method / architecture	Dataset / evaluation	Key findings
P1	[Author, year, title, venue, DOI]	[Problem]	[Method]	[Dataset/metrics]	[Findings]
P2	[Citation]	[Problem]	[Method]	[Dataset/metrics]	[Findings]
P3–P15	[Add one row for each remaining paper]	[Complete]	[Complete]	[Complete]	[Complete]

2.1 Synthesis and Comparison

[Compare the papers by approach, cloud services, performance measures, security, scalability, cost, and limitations. Identify common trends and conflicting findings.]

2.2 Individual Research Gap Analysis

Each student must prepare an independent analysis for their assigned papers. Do not copy a paper's stated research gap; explain the gap in your own words and support it with evidence from the comparison.

Paper ID	Existing method	Advantages	Limitations	Research gap identified by student	Possible improvement
[P1]	[Method]	[Advantages]	[Limitations]	[Your analysis]	[Improvement]
[P2]	[Method]	[Advantages]	[Limitations]	[Your analysis]	[Improvement]
[Add rows as required]					

2.3 Consolidated Research Gap

[Summarise the main gap that the proposed project will address and explain how the reviewed literature supports this conclusion.]

3. Project Objectives

Write 4–6 specific, measurable, achievable objectives.

1. Objective 1: [State an objective and, where possible, a measurable success criterion.]
2. Objective 2: [State an objective and, where possible, a measurable success criterion.]
3. Objective 3: [State an objective and, where possible, a measurable success criterion.]
4. Objective 4: [State an objective and, where possible, a measurable success criterion.]
5. Objective 5: [State an objective and, where possible, a measurable success criterion.]
6. Objective 6: [State an objective and, where possible, a measurable success criterion.]

3.1 Expected Outcomes

[List expected deliverables and how each objective will be assessed.]

4. Novelty Summary

Limit this section to approximately one page. Clearly distinguish the proposed contribution from features already available in existing systems.

Novelty dimension	Existing approach / limitation	Proposed difference	How it will be demonstrated
New feature	[Existing capability]	[New feature]	[Test/evidence]
Algorithm / processing	[Baseline]	[Proposed method]	[Comparison metric]
Cloud architecture / AWS integration	[Existing design]	[Architecture improvement]	[Load, latency, cost or reliability test]
Security / automation / scalability	[Current limitation]	[Proposed improvement]	[Verification method]

[State the central novelty claim in 3–5 clear sentences. Avoid unsupported claims such as 'first ever' unless a systematic search substantiates them.]

5. Proposed System and Architecture

5.1 System Overview

[Describe the users, inputs, processing steps, outputs, and the main system components.]

5.2 Diagram 1 – AWS Cloud Architecture (Mandatory)

[Insert the AWS architecture diagram here. Label all AWS services and arrows. Show data flow, storage, processing, authentication/authorisation, notifications, monitoring, and external users/data sources where relevant.]

Figure 1. Proposed AWS Cloud Architecture.

[Explain the role of each component and describe the end-to-end data flow.]

5.3 Diagram 2 – Complete System Architecture (Mandatory)

[Insert the complete project/system workflow diagram here. Show the user/application layer, input sources, core logic or ML components if applicable, database, AWS layer, outputs, and feedback/monitoring flow.]

Figure 2. Complete System Architecture.

[Explain how the complete workflow operates, from input to output.]

5.4 Workflow / Use Cases

[Add a numbered workflow, flowchart, use-case diagram, or sequence diagram as appropriate.]

5.5 Security, Availability and Scalability Design

[Explain IAM least privilege, authentication, encryption, network controls, backups, monitoring, fault handling, and scaling choices applicable to the project.]

6. Dataset Description

Complete this section if the project uses a dataset. If the project uses live data, synthetic data, or no dataset, describe that choice and its implications.

Dataset attribute	Details
Dataset name	[Name]
Source / organisation	[Provider]
URL	[Direct URL]
Dataset size	[MB/GB]
Number of records / samples	[Count]
Number of features	[Count]
Data types / format	[CSV, JSON, images, logs, etc.]
License / usage restrictions	[License and conditions]
Purpose in this project	[Training, testing, monitoring, storage, etc.]
Preprocessing required	[Cleaning, missing values, normalisation, encoding, splitting, etc.]
Privacy / ethical considerations	[Sensitive fields, anonymisation, consent, retention, etc.]

6.1 Data Preparation Plan

[Describe train/validation/test split where applicable, preprocessing steps, and how data leakage will be avoided.]

7. AWS Services Planning

List only services relevant to this project. Mark each service as Planned, Configured, Tested, or Deployed. Include cost and security considerations.

AWS service	Purpose in this	Input/output	Status	Security / cost

	project	or integration		considerations
Amazon S3	[Object/file/dataset storage]	[What writes/reads data]	[Planned/Configured/Tested/Deployed]	[Access, encryption, storage class]
Amazon EC2 / AWS Lambda	[Compute or event processing]	[Integration]	[Status]	[Runtime, scaling, cost]
Amazon RDS / database service	[Structured data storage, if required]	[Integration]	[Status]	[Backups, access, cost]
Amazon API Gateway	[API entry point, if required]	[Integration]	[Status]	[Authentication, throttling]
AWS IAM / Amazon Cognito	[Permissions / user authentication, if required]	[Integration]	[Status]	[Least privilege, MFA]
Amazon CloudWatch	[Logs, metrics, alarms]	[Integration]	[Status]	[Retention, alert thresholds]
Amazon SNS	[Notifications, if required]	[Integration]	[Status]	[Access policy, delivery]
Other service(s)	[Explain why required]	[Integration]	[Status]	[Considerations]

7.1 AWS Deployment and Network Design

[Describe region, VPC/subnets if used, security groups, endpoints, environment configuration, deployment method, and access boundaries.]

7.2 Cost Estimation

[Provide an estimated monthly cost, assumptions, usage levels, pricing source/date, and cost-control measures. Clearly label estimates; do not present them as actual bills.]

8. Methodology and Implementation Plan

8.1 Overall Methodology

[Describe the phases: requirements, literature review, data acquisition, design, implementation, integration, testing, and evaluation.]

8.2 Tools and Technologies

Tool / technology	Purpose	Version / configuration
AWS services	[Purpose]	[Region / configuration]
Programming language / framework	[Purpose]	[Version]
Database / ML framework / other	[Purpose]	[Version]

tools		
Git and GitHub	Version control and collaboration	[Repository / branch details]

8.3 Step-by-Step Implementation Plan

7. Step 1 – Requirements and success criteria: [Details]
8. Step 2 – Dataset or input preparation: [Details]
9. Step 3 – Architecture and AWS resource setup: [Details]
10. Step 4 – Application / backend / model development: [Details]
11. Step 5 – Integration and security configuration: [Details]
12. Step 6 – Testing and performance evaluation: [Details]
13. Step 7 – Deployment, monitoring, and documentation: [Details]

8.4 Evaluation Metrics

[Specify relevant measures such as response time, throughput, availability, scalability, cost, accuracy/precision/recall, or error rate. Define how each metric is measured.]

9. Implementation / Development

For Phase I, document completed work and planned work separately. Add dated screenshots with figure numbers and captions; do not include secrets, access keys, passwords, tokens, or personal data.

9.1 Implemented Modules

Module	Implementation details	AWS services / tools	Evidence (figure, commit, PR)	Status
[Module 1]	[What was implemented]	[Services/tools]	[Evidence]	[Status]
[Module 2]	[Details]	[Services/tools]	[Evidence]	[Status]
[Module 3]	[Details]	[Services/tools]	[Evidence]	[Status]

9.2 Screenshots and Configuration Evidence

[Insert relevant console, application, deployment, API, database, or monitoring screenshots with short explanations.]

9.3 GitHub Repository

[Repository name should follow the guideline format: ProjectName_Cloud_Project_2026. Add the repository URL and briefly explain the folders and key files.]

Suggested structure:

- README.md, LICENSE, .gitignore
- docs/ – project report, literature survey, research gap, objectives, novelty
- architecture/ – AWS architecture, system architecture, workflow diagrams
- dataset/ – raw and processed data where licensing permits, plus dataset description
- src/ – frontend, backend, ML model, and AWS-related code as applicable
- results/ – graphs, screenshots, evaluation outputs
- presentation/ – review slides

10. Testing, Results and Observations

10.1 Test Plan and Test Cases

Test ID	Scenario	Input / setup	Expected result	Observed result	Pass/Fail
T01	[Functional test]	[Input/setup]	[Expected]	[Actual]	[Status]
T02	[Security/access test]	[Input/setup]	[Expected]	[Actual]	[Status]
T03	[Performance/scalability test]	[Input/setup]	[Expected]	[Actual]	[Status]

10.2 Results

[Insert actual results using tables, graphs, screenshots, and measured values. State environment, test conditions, sample size, and repetitions where relevant.]

10.3 Baseline Comparison

[Compare the proposed system with a justified baseline or existing approach using the same conditions and metrics. If not yet evaluated, state 'not yet evaluated' and provide the planned experiment.]

10.4 Results Summary

[Summarise which objectives have been achieved and cite the evidence supporting each claim.]

11. Discussion, Limitations and Risk Analysis

11.1 Discussion

[Interpret the findings and explain why observed results matter.]

11.2 Limitations

[Document technical, dataset, cost, time, security, or evaluation limitations.]

11.3 Risks and Mitigation

Risk	Likelihood / impact	Mitigation	Residual risk
Unexpected AWS cost	[Level]	[Budget alerts, shut down unused resources, least-cost design]	[Level]
Credential exposure / misconfiguration	[Level]	[IAM least privilege, secret management, no keys in GitHub]	[Level]
Dataset quality / availability	[Level]	[Validation, backup, alternative source]	[Level]
Other project-specific risk	[Level]	[Mitigation]	[Level]

12. Team Contribution and GitHub Management

For team projects, clearly document each member's individual work. For an individual project, mark the other columns as Not Applicable.

Activity	Student 1	Student 2	Student 3
Literature survey and assigned paper analysis	[Contribution]	[Contribution]	[Contribution]
Research gap and novelty	[Contribution]	[Contribution]	[Contribution]
Frontend / application	[Contribution]	[Contribution]	[Contribution]
Backend / APIs / database	[Contribution]	[Contribution]	[Contribution]
AWS services and security	[Contribution]	[Contribution]	[Contribution]
Dataset / AI-ML (if applicable)	[Contribution]	[Contribution]	[Contribution]
Testing and results	[Contribution]	[Contribution]	[Contribution]
Documentation and presentation	[Contribution]	[Contribution]	[Contribution]

12.1 Branch and Pull Request Workflow

[Document main, develop, and feature/studentX branches; commit and pull request practices; code review; merge-conflict resolution; and final release/tag.]

12.2 Contribution Evidence

[Add each student's GitHub profile/username, branch, key commits, pull requests, code reviews, and other verifiable contributions. The guideline expects approximately 20–30 meaningful commits per student, at least two pull requests, code-review participation, and regular weekly commits.]

13. Conclusion and Future Enhancements

13.1 Conclusion

[Summarise the problem, proposed cloud solution, work completed, key verified findings, and extent to which objectives were met.]

13.2 Future Enhancements

[List realistic next steps such as broader testing, stronger security controls, improved automation, additional AWS services, or larger-scale evaluation.]

References

Use APA style consistently for research papers, books, datasets, AWS documentation, and other sources. Include DOI or stable URL where available. Cite each source in the body of the report.

[1] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[2] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[3] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[4] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[5] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[6] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[7] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[8] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[9] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[10] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[11] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[12] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[13] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[14] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[15] [Author(s). (Year). Title of paper. Journal/Conference, volume(issue), page range. https://doi.org/xxxxx]

[Add additional references for AWS documentation, datasets, libraries, and tools used.]

Appendix A – Project Timeline

Phase / task	Planned start	Planned end	Owner(s)	Status / evidence
Literature survey	[Date]	[Date]	[Name(s)]	[Status]
Research gap and novelty	[Date]	[Date]	[Name(s)]	[Status]
Architecture design	[Date]	[Date]	[Name(s)]	[Status]
Implementation and integration	[Date]	[Date]	[Name(s)]	[Status]
Testing and report	[Date]	[Date]	[Name(s)]	[Status]

Appendix B – Review Checklist

☐ Abstract is 200–300 words and describes the problem, challenges, proposed solution, AWS services, and status/results.

☐ 15 research papers are included, with citations and a comparative literature survey.

☐ Each student has completed their own research gap analysis for assigned papers.

☐ 4–6 measurable objectives are listed.

☐ Novelty summary is limited to about one page and is supported by comparison with existing work.

☐ Both required architecture diagrams are included and explained.

☐ Dataset name, source, URL, size, record count, feature count, data type, license, purpose, and preprocessing are documented where applicable.

☐ AWS services, their purpose, status, security considerations, and estimated cost are documented.

☐ Implementation evidence, test cases, actual results, limitations, and future work are included.

☐ GitHub repository, branch workflow, pull requests, commits, and individual contributions are documented.

☐ References use APA style; all borrowed ideas, figures, datasets, and documentation are cited.

☐ No credentials, passwords, tokens, or confidential data appear in the report or repository.

☐ Report follows the specified font, spacing, margin, heading, and page-number formatting.